

Does Distractor Position Affect LLM Reasoning?

An Exploratory Study of Irrelevant Context in Mathematical Reasoning

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Research Question

Does the position of irrelevant information within a prompt affect the reasoning accuracy of a large language model?

Hypothesis

I expect irrelevant information to affect model performance differently depending on where it appears in the prompt. In particular, I expect distractors placed in the middle of the problem to be more disruptive because they interrupt the flow of information needed for reasoning.

Experimental Conditions

1. Clean Condition

The original reasoning problem with no irrelevant information added.

2. Distractor First

The same irrelevant sentence is placed before the information needed to solve the problem.

3. Distractor Middle

The irrelevant sentence is inserted between pieces of information needed to solve the problem.

4. Distractor Last

The irrelevant sentence is placed after all information needed to solve the problem but immediately before the final question.

Example

Clean:

Selin buys 4 cups of latte for €3 each. How much does she spend?

Distractor First:

Selin's favorite color is pink. Selin buys 4 cups of latte for €3 each. How much does she spend?

Distractor Middle:

Selin buys 4 cups of latte. Selin's favorite color is pink. Each cup of latte costs €3. How much does she spend?

Distractor Last:

Selin buys 4 cups of latte for €3 each. Selin's favorite color is pink. How much does she spend?

Independent Variable

Position of irrelevant information:

- Clean
- Beginning
- Middle
- End

Dependent Variable

Answer accuracy:

- Correct answer = 1
- Incorrect answer = 0

Sample

30 reasoning problems $\times$ 4 conditions = **120 total prompts**

Goal

The experiment tested whether the same irrelevant information became more or less disruptive depending only on where it appeared in the prompt.

Dataset

This experiment used 30 multi-step mathematical reasoning problems sampled from the GSM8K benchmark (Cobbe et al., 2021). The same problems were presented under four conditions: clean, distractor first, distractor middle, and distractor last. The distractor content remained constant across the three distractor conditions for each problem so that its position was the primary experimental manipulation.

Selection Criteria

- Multi-step reasoning required
- Clear numerical ground-truth answer
- Problems were required to allow an irrelevant sentence to be inserted naturally at different positions
- The original mathematical information and correct answer were required to remain unchanged across all four conditions

Sample Selection

An initial sample of 30 problems was drawn reproducibly from the GSM8K test set using a fixed random seed (random_state = 42). Before any model testing was conducted, three problems were excluded because their wording or answer interpretation could introduce ambiguity unrelated to the experimental manipulation. Three replacement problems were then sampled from the remaining test set using a second fixed random seed (random_state = 43). The resulting 30-question sample was fixed before the experimental conditions were generated.

Distractor Generation Protocol

Each reasoning problem was assigned one irrelevant distractor sentence. The same distractor was used across the First, Middle, and Last conditions for that problem, with only its position changing.

Each distractor was required to:

- consist of one short sentence;
- be related naturally to a person, object, or setting already mentioned in the problem;
- contain no numerical information;
- introduce no information required to solve the problem;
- not contradict any information in the original question;
- avoid mathematical or comparative language;
- remain identical across all three distractor conditions.

Pilot Study

A preliminary pilot was conducted using Qwen2.5-3B-Instruct. The model achieved only 2/30 correct answers in the clean condition, creating a floor effect that made it unsuitable for evaluating differences between distractor positions. The pilot was therefore excluded from the main analysis, and a larger model was selected before conducting the final experiment.

Main Experiment

The main experiment was conducted using Qwen2.5-7B-Instruct (Qwen Team et al., 2024). Each of the 30 selected problems was presented under all four experimental conditions, resulting in 120 total prompts. The model was instructed to solve each problem step by step and provide its final numerical answer in a standardized format. Generation was deterministic (`do_sample = False`) to keep the model settings consistent across conditions. Each response was then compared with the GSM8K ground-truth answer and coded as either correct (1) or incorrect (0).

Technical Settings

The model was run using 4-bit quantization, with a maximum generation length of 768 tokens. The longer generation limit was used to ensure that the model had enough space to complete its reasoning and provide a final answer without responses being truncated. The same generation settings were used across all four experimental conditions.

Results

Qwen2.5-7B-Instruct answered 28 out of 30 problems correctly in the clean condition, resulting in an accuracy of 93.3%. Accuracy was 96.7% when the distractor appeared first, 100% when it appeared in the middle, and 96.7% when it appeared last. These results did not support my initial hypothesis that distractors placed in the middle would be the most disruptive. In absolute terms, however, this difference corresponded to only two additional correct answers out of 30.

Since the same 30 problems were evaluated under all four conditions, I used Cochran's Q test to compare the paired binary outcomes. The difference in accuracy across the four conditions was not statistically significant ($Q = 4.00$, $p = .261$). Therefore, the experiment did not provide sufficient evidence that distractor position systematically affected overall answer accuracy.

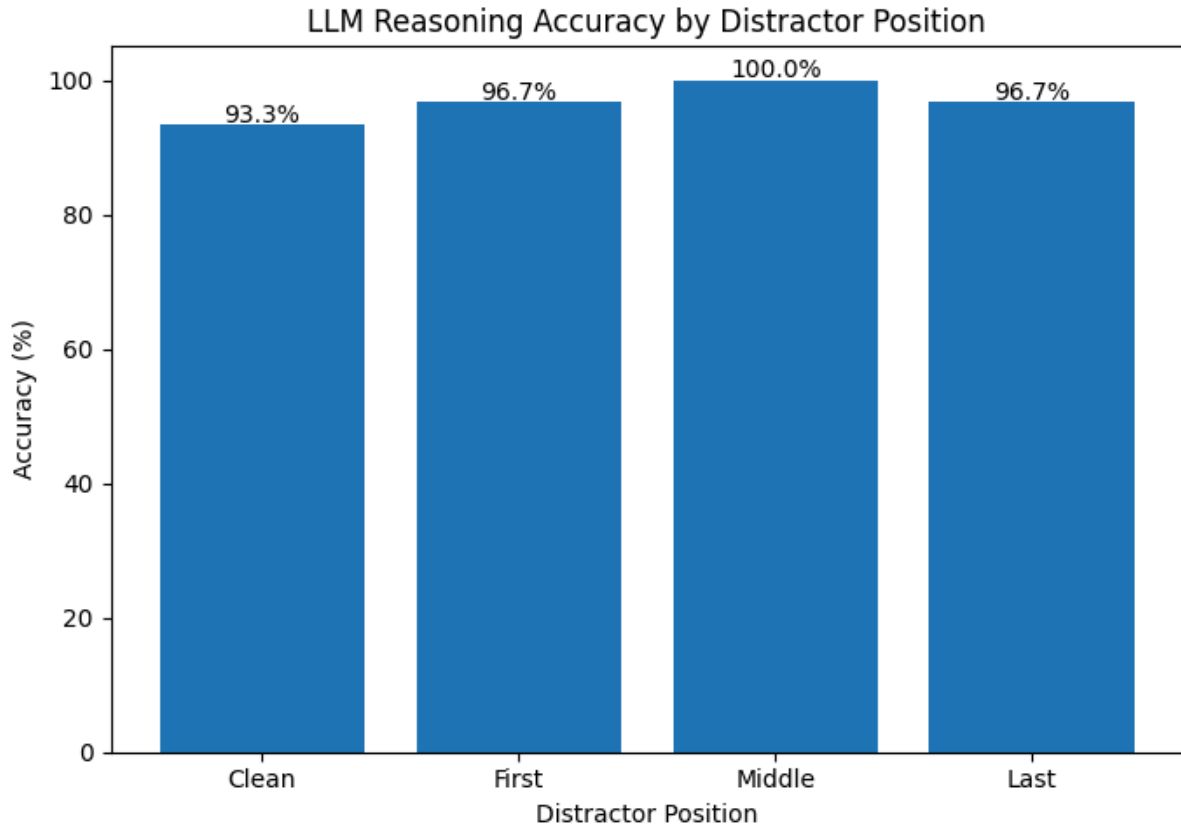


Figure 1. *Qwen2.5-7B-Instruct* answer accuracy across distractor-position conditions ($n = 30$ problems per condition)

Discussion

I initially expected the middle distractor to interfere with the model's reasoning the most, but the results did not support this. Accuracy remained very similar across all four conditions, and the differences were not statistically significant. Although the middle condition reached 100% accuracy, I do not think this suggests that placing irrelevant information in the middle actually improves reasoning. With only 30 questions, the difference between 93.3% and 100% represents just two answers, which is too small to make a broader conclusion.

What I found more interesting was that the model did not always reason in the same way when irrelevant information was added. In Question 10, for example, the model forgot to include Sara's initial \$10 savings in the clean condition and reached the wrong answer. In all three distractor conditions, it correctly included the same \$10 and solved the problem. The distractor itself had nothing to do with the calculation, yet its presence changed the reasoning path the model followed.

Question 28 showed an even more notable pattern. In the clean, first, and last conditions, the model calculated the required average across six tests, even though the lowest test was supposed to be removed. Only in the middle condition did it correctly recognize that the average should be calculated across five tests, producing the correct answer of 98 instead of 191. Again, the distractor provided no useful information for solving the problem.

These examples do not show that irrelevant information helps an LLM reason better. However, they made me question how stable the model's reasoning actually is. If the information required to solve a problem remains exactly the same, but an unrelated sentence can occasionally change the reasoning path and final answer, then seemingly minor changes in a prompt may matter in ways that are difficult to predict.

Limitations

This experiment has several limitations. First, the sample contained only 30 problems, meaning that a single response changed overall accuracy by more than three percentage points. Second, only one language model was tested. The model also performed very well in the clean condition, creating a possible ceiling effect and leaving relatively few errors for the distractors to influence. Finally, although each problem used the same distractor across its three distractor conditions, the distractor sentences differed between problems. A larger study could use more problems, multiple models, and more difficult reasoning tasks. Future work could also examine whether the type of irrelevant information matters in addition to its position.

Conclusion

In this small exploratory experiment, I did not find evidence that distractor position systematically affected LLM reasoning accuracy. However, the analysis of individual responses showed that irrelevant context could occasionally change how the model approached the same problem. This raised a broader question that I found more interesting than my original hypothesis: how stable is an LLM's reasoning when the useful information stays the same, but irrelevant context around it changes?

References

- Cobbe, K., Kosaraju, V., Bavarian, M., Chen, M., Jun, H., Kaiser, L., Plappert, M., Tworek, J., Hilton, J., Nakano, R., Hesse, C., & Schulman, J. (2021). *Training verifiers to solve math word problems*. arXiv:2110.14168.
- Qwen Team, Yang, A., Yang, B., Zhang, B., Hui, B., Zheng, B., Yu, B., et al. (2024). *Qwen2.5 technical report*. arXiv:2412.15115.